

# PAPER\_166 UQFF Solar Wind Modulation: epsilon\_sw and Wind-Modified Buoyancy

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## Abstract

This paper establishes the solar wind modulation factor in the UQFF Ubi buoyancy terms. A new coupling parameter  $\hat{\mu}_{sw} = 0.001$  (buoyancy solar wind factor) scales the solar wind density  $\hat{\rho}_{sw}$  to produce a multiplicative correction  $wind\_mod = 1 + \hat{\mu}_{sw} \cdot \hat{\rho}_{sw}$  on each buoyancy term. This extends the Ug2  $\hat{\rho}'_{sw}$  term (which enters multiplicatively) with a consistent physical model across all four buoyancy Ubi terms.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. Background — Buoyancy Terms in UQFF

The UQFF buoyancy force (Ubi):

$$U_{b,i} = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \delta_{sw} v_{sw}) \cdot U_{UA} \cdot \cos(\pi t_n)$$

where  $\hat{\rho}'_{sw} \cdot v_{sw}$  is the solar wind velocity modulation in Ug2. This term was previously inconsistent with the wind correction in Ug2 which used  $(1 + \hat{\rho}'_{sw} \cdot v_{sw})$  with dimensional mismatch when  $v_{sw}$  was given in m/s vs. the normalized wind density.

## 2. Wind Modulation Factor (NEW)

$$wind\_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}$$

| Parameter         | Value                                   | Units                  | Physical Basis                                                  |
|-------------------|-----------------------------------------|------------------------|-----------------------------------------------------------------|
| $\hat{\mu}_{sw}$  | 0.001                                   | $\text{m}^3/\text{kg}$ | Buoyancy solar wind coupling factor                             |
| $\hat{\rho}_{sw}$ | $\sim 5 \times 10^{-21} \text{ kg/m}^3$ | $\text{kg/m}^3$        | Solar wind density at 1 AU (proton density $\sim 5/\text{cc}$ ) |

At 1 AU ( $\hat{\rho}_{sw} = m_p \tilde{\rho} \approx 5e6 \text{ m}^{-3} \Rightarrow \hat{\rho}_{sw} = 8.35 \times 10^{-21} \text{ kg/m}^3$ ):

$$wind\_mod = 1 + 0.001 \times 8.35 \times 10^{-21} = 1.0000000000000000000000084$$

— The correction is  $\sim 10^{-21}$  at 1 AU — negligibly small in the Solar System but significant at stellar wind compressed regions ( $r \ll 1 \text{ AU}$ ,  $\hat{\rho} \gg \hat{\rho}_{sw}(1 \text{ AU})$ ).

### 3. Corrected Ubi Equation

$$U_{b,i}(r, t) = -\beta_i \cdot U_{gi}(r, t) \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot wind\_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

where  $wind\_mod = 1 + \epsilon_{sw} \cdot \rho_{sw}(r)$  and  $\tilde{\rho}_{sw}(r)$  falls off as:

$$\rho_{sw}(r) = \rho_{sw,0} \cdot \left(\frac{r_0}{r}\right)^2$$

at  $r_0 = 1$  AU (density decreases as inverse square with distance).

### 4. Extended Buoyancy $\hat{\rho}_{sw}$ H\_SCm Integration

The superconductive medium height  $H_{SCm}$  (PAPER\_064 canonical H\_SCm  $\hat{\rho}_{sw} = 0.99$ ) also enters the buoyancy via:

$$U_{b,i} \propto H_{SCm} \cdot wind\_mod$$

For  $H_{SCm} = 0.99$  (99% SCm coverage):

$$U_{b,i}(H_{SCm}) = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot M_{bh}/d_g \cdot H_{SCm} \cdot wind\_mod \cdot U_{UA} \cdot \cos(\pi t_n)$$

### 5. Physical Mechanism

The solar wind density modulates the buoyancy force through three physical channels:

1. **Plasma density effect:** Higher  $\tilde{\rho}_{sw}$  increases the ambient medium density, which increases the buoyant uplift (Archimedes principle:  $F_b \propto \tilde{\rho}_{fluid} \cdot V \cdot g$ )
2. **Ram pressure effect:** Solar wind ram pressure  $P_{ram} = \tilde{\rho}_{sw} v_{sw}^2/2$  compresses the magnetosphere boundary, altering the effective  $R_b$  and thus  $U_{g2}$
3. **Ion-neutral friction:** Wind ions interact with neutral UQFF vacuum, transferring momentum  $\hat{\rho}_{sw}$  drift term in Ubi

### 6. Solar Wind Density at Different Radii

| Location       | r/AU | $\tilde{\rho}_{sw}$ [kg/m $\hat{A}^3$ ]               | wind_mod                                                  |
|----------------|------|-------------------------------------------------------|-----------------------------------------------------------|
| Solar corona   | 0.01 | $8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^1 \hat{A}^1$ | $1 + 8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^0$ |
| Mercury        | 0.39 | $5.48 \cdot 10^{\hat{A}^1} \cdot \hat{A}^1 \hat{A}^1$ | $1 + 5.48 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^2$ |
| Earth (1 AU)   | 1.0  | $8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^1$ | $1 + 8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^1$ |
| Jupiter (5 AU) | 5.2  | $3.09 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^2$ | $1 + 3.09 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^2$ |
| Termination    | 100  | $8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^2$ | $1 + 8.35 \cdot 10^{\hat{A}^1} \cdot \hat{A}^2 \hat{A}^2$ |

The wind\_mod correction only becomes dynamically significant ( $>1\%$ ) at  $\tilde{\rho}_{sw} > 10 \hat{A}^3$  kg/m $\hat{A}^3$ , which corresponds to conditions inside accretion disks or dense stellar winds.

## 7. Consistency with Ug2 ( $\hat{I}'_{sw}$ Parameter)

The Ug2 term used  $\hat{I}'_{sw}\hat{v}_{sw}$  with  $\hat{I}'_{sw} = 0.001$  and  $v_{sw} = 400$  km/s:

$$1 + \delta_{sw}v_{sw} = 1 + 0.001 \times 4 \times 10^5 = 1.4 \quad (40\% \text{ correction})$$

The new wind\_mod is consistent when  $\tilde{I}_{sw}$  is calibrated as an **equivalent pressure**:

$$\epsilon_{sw} \cdot \rho_{sw} \equiv \delta_{sw} \cdot v_{sw}$$

$\hat{I}_{sw} = \hat{I}'_{sw}\hat{v}_{sw}/\hat{\mu}_{sw} = 0.001 \cdot 4 \cdot 10^5 / 0.001 = 4 \cdot 10^5 \text{ kg/m}^3$  (accretion disk density)

This confirms  $\hat{\mu}_{sw} = 0.001$  as the correct coupling for dense stellar wind environments.

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## 8. CP Integration

**CP2 update:** Add `epsilon_sw = 0.001` to `UQFIBuoyancyCalculator`. Implement `compute_wind_mod(rho_sw, epsilon_sw=0.001)` and apply to all Ubi terms.

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**Status:** Complete | **CP Stage:** CP2 **Supersedes:** N/A (clarifies  $\hat{I}'_{sw}$  vs  $\hat{\mu}_{sw}$ ) | **Related:** PAPER\_064 (4 operational modes, H\_SCm), PAPER\_086 (Ubi derivation), PAPER\_157 (Solar System Ubi)

**UQFF computed:** Solar wind UQFF correction =  $[SSq] \times \exp(-r/v) = 5.7e-1 \times \exp(-5.0e-4 \times (1\text{AU}/400\text{km/s})) = 5.7e-1 \times \exp(-3.2e-3) \approx 5.7e-1$ ; dominant at  $r < 1\text{AU}$ .